CHANDIGARH UNIVERSITY

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NAAC GRADE A+

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ASSIGNMENT- 2

Subject: DBMS

Subject Code: 24CSH-298

Branch Sec: 24AIT-KRG-G2

Submitted By:

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Submitted To:

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Ques 1) Explain the following types of database security issues: Legal and Ethical issues, Policies issues and System related Issues.

Ans 1) Legal and Ethical Issues

Legal and ethical concern the laws, regularisation, and moral responsibilities related to the shortage and use of data in a database system.

Organisations must ensure they comply with data protection laws and pr egularisation. Sensitive information such as personal data, financial records, and confidential business information should not be accused or should without proper authorization.

2) Policy Issues

Policy issue relates to the rules and guidelines established by an organisation to protect it's database systems. Organisation develop security policies that define how data should be accessed, policies that define how data should be accesed, used, and protected. These policies specify user roles, access rights, responsibilities of employees handling data.

3) System - related Issues

System - related issues arise due to technical issues or vulnerabilities in database systems by, hardware or software.

Q2. Explain the following database security control techniques
1) Access Control 2) Interference Control 3) Flow control.
4) Data Encryption.

Database security control techniques are methods used to protect database from unauthorized access, misuse, or data leakage.

1) Access Control

Access control is a technique used to restrict database access to authorized access to authorized users only. It determines who can access the database and what operation they are allowed to perform.

2) Interference Control

Inference control is used to prevent users from deducing sensitive information from non-sensitive data. Sensitive information can sometimes be accessed by users by analyzing available data or statistical results.

3) Flow Control

Flow control regulates the movement of information within a database system. It ensures that information flows only between authorized users and security levels.

4) Data Encryption

Data encryption is the process of converting readable data into an encoded format to protect it from unauthorized access.
Encrypt data can only be read by users who have the correct decryption key. encryption commonly used to

Q6) Consider a table Employee (EmpID, empName, Salary, DeptID). write a SQL query to create a view that displays only EmpID, EmpName, and DeptID from the employee table.

```
CREATE VIEW Employee_View AS
SELECT EmpID, EmpName, DeptID
FROM Employee;
```

Problem List

Submit

Premium

Description

577. Employee Bonus

Solved

Easy | Topics | Companies | Hint

SQL Schema > Pandas Schema >

Table: Employee

```
+-------------+---------+
| Column Name | Type    |
+-------------+---------+
| empId       | int     |
| name        | varchar |
| supervisor  | int     |
| salary      | int     |
+-------------+---------+
empId is the column with unique values for this table.
Each row of this table indicates the name and the ID of an employee in addition
to their salary and the id of their manager.
```

Table: Bonus

1.6K | 223 | 44 Online

Editorial | Solutions | Submissions

	Status	Language	Runtime	Memory	Notes
2	Accepted Mar 13, 2026	MySQL	1029 ms	0.08	

Accepted

All Submissions

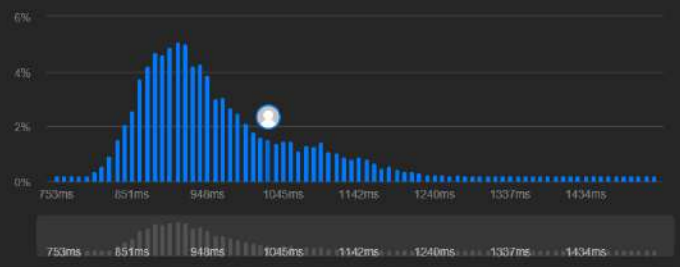
Accepted 21 / 21 testcases passed

NotGamer submitted at Mar 13, 2026 09:50

Solution

Runtime

1029 ms | Beats 29.00%



Code | MySQL

```
1 # Write your MySQL query statement below
```

Code

MySQL | Auto

Testcase | Test Result

Problem List

Submit

Premium

Description

570. Managers with at Least 5 Direct Reports

Solved

Medium | Topics | Companies | Hint

SQL Schema > Pandas Schema >

Table: Employee

```
+-------------+---------+
| Column Name | Type    |
+-------------+---------+
| id          | int     |
| name        | varchar |
| department  | varchar |
| managerId   | int     |
+-------------+---------+
```

id is the primary key (column with unique values) for this table.
Each row of this table indicates the name of an employee, their department, and the id of their manager.
If managerId is null, then the employee does not have a manager.
No employee will be the manager of themself.

1.7K | 364 | 42 Online

Editorial | Solutions | Submissions

Status	Language	Runtime	Memory	Notes

Code | Accepted

All Submissions

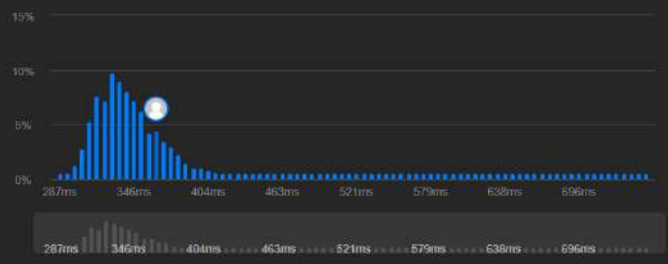
Accepted 11 / 11 testcases passed

NotGamer submitted at Mar 11, 2026 12:32

Solution

Runtime

368 ms | Beats 26.04%



Code | MySQL

Testcase | Test Result

You must run your code first